

Sr. No	Question Content	Option 1	Option 2	Option 3	Option 4	Correct
1	What does OSI stand for?	Open Systems Interconnection	Open Standard Integration	Open System Interlink	Open Systems Interface	Open Systems Interconnection
2	Which topology connects all devices to a single central hub?	Star	Mesh	Ring	Bus	Star
3	What are the number of cable links required for n devices connected in mesh topology?	n-1	n	$n(n-1)/2$	1	$n(n-1)/2$
4	Which topology is characterized by each device being connected to every other device?	Star	Ring	Mesh	Bus	Mesh
5	What is the main function of a router?	Store data	Forward packets	Encrypt data	Manage user sessions	Forward packets
6	Which of these devices operate on network layer of OSI reference model?	Repeater	Gateway	Modem	Router	Router
7	In which type of multiplexing is time divided among users?	Frequency Division	Wavelength Division	Time Division	Space Division	Time Division
8	What is the maximum length of a twisted pair cable for Ethernet?	100 meters	200 meters	300 meters	400 meters	100 meters
9	Which layer of the OSI model does error detection and correction occur?	Application Layer	Data Link Layer	Network Layer	Physical Layer	Data Link Layer
10	Which protocol is not used for error detection ?	ARP	Checksum	CRC	Parity bit	ARP
11	What does TCP stand for?	Transmission Control Protocol	Transfer Control Protocol	Time Control Protocol	Transmit Control Protocol	Transmission Control Protocol
12	Which of the following is a connection oriented and reliable protocol?	TCP	UDP	ICMP	FTP	TCP
13	Which of the following is a connectionless protocol?	TCP	UDP	ICMP	FTP	UDP
14	What is the purpose of the DHCP protocol?	Assign IP address	Encrypt data	Manage user session	Store files	Assign IP addresses
15	Which media type is immune to electromagnetic interference?	Twisted pair	Coaxial cable	Fiber optic	Copper cable	Fiber optic
16	What does a firewall do?	Encrypts data	Filters traffic	Stores data	Connects networks	Filters traffic
17	What are the number of cable links required for n devices connected in star topology?	n-1	n	$n(n-1)/2$	1	n
18	What protocol is used to resolve IP addresses to MAC addresses?	TCP	ARP	DNS	ICMP	ARP
19	Which of the following is not a characteristic of IPv6?	Increased address space	Simplified header	Compatibility with IPv4	128-bit address	Compatibility with IPv4
20	What type of network covers a large geographical area?	LAN	WAN	PAN	MAN	WAN
21	Which of the following is a function of the transport layer?	Routing packets	Data encryption	Ensuring reliable delivery	Network addressing	Ensuring reliable delivery
22	Which error detection method uses a polynomial division?	Parity Check	Checksum	CRC	Bit Check	CRC
23	What is the primary purpose of a VPN?	File sharing	Secure remote access	Video streaming	Data backup	Secure remote access

24	What does CSMA/CD stand for?	Controlled Access with Collision Detection	Carrier Sense Multiple Access with Collision Detection	Common Sense Media Access	Carrier Signal Media Access	Carrier Sense Multiple Access with Collision Detection
25	Which layer of the OSI model is responsible to controls the dialogues?	Network Layer	Transport Layer	Session Layer	Data Link Layer	Session Layer
26	What is the function of the DNS?	Manages user sessions	Encrypts data	Filters traffic	Translates IP addresses	Translates IP addresses
27	What is the transmission time for a packet of size 20 Mb sent by a station using a channel of 250 kbps speed?	300 seconds	20 seconds	80 seconds	40 seconds	80 seconds
28	What type of network is a collection of devices connected together in one physical location, such as a building, office, or home.?	WAN	LAN	PAN	MAN	LAN
29	Which protocol is used for email transmission?	FTP	SMTP	HTTP	SNMP	SMTP
30	In a star topology, what happens if the central hub fails?	All devices are unaffected	Only the hub fails	All devices lose connectivity	Only some devices lose connection	All devices lose connectivity
31	What does the term "bandwidth" refer to in networking?	Data packet size	Transmission capacity	Latency	Protocol type	Transmission capacity
32	Which of the following protocols is used for secure web communication?	FTP	HTTP	HTTPS	SMTP	HTTPS
33	What is the function of a switch in a network?	Connects multiple networks	Filters packets	Connects devices within a network	Encrypts data	Connects devices within a network
34	Which device operates at the data link layer of the OSI model?	Router	Switch	Hub	Firewall	Switch
35	Which of the following methods is used for access control in networks?	CSMA/CD	Token Ring	ALOHA	All of the above	All of the above
36	What is the purpose of network addressing?	Identify network paths	Locate devices	Secure communications	Manage traffic	Locate devices
37	Which layer of the OSI model handles encryption and decryption?	Application Layer	Presentation Layer	Session Layer	Transport Layer	Presentation Layer
38	What does the acronym "IP" stand for in networking?	Interconnected Protocol	Internal Protocol	Internet Protocol	International Protocol	Internet Protocol
39	Which of the following is a disadvantage of a bus topology?	Difficult to add devices	Single point of failure	High cost	Complexity	Single point of failure
40	Which technology allows multiple devices to share a single communication channel?	Multiplexing	Switching	Routing	Addressing	Multiplexing
41	What is the main purpose of Network address translation?	Enhance security	Conserve IP addresses	Improve speed	Simplify routing	Conserve IP addresses
42	Which protocol is used for network management?	SNMP	SMTP	DHCP	FTP	SNMP
43	What does the term "latency" refer to in networking?	Data packet size	Delay in communication	Number of users	Network speed	Delay in communication
44	What is a key benefit of using fiber optic cables?	High speed	Low cost	Easy installation	Short distance	High speed
45	Which topology is the most fault-tolerant?	Bus	Star	Mesh	Ring	Mesh
46	What layer of the OSI model does IP operate on?	Network Layer	Data Link Layer	Transport Layer	Application Layer	Network Layer
47	Which of the following protocols is primarily responsible for transferring files over the internet?	FTP	SMTP	SNMP	HTTP	FTP

48	What is the primary function of an Ethernet switch?	Manage the flow of data packets	Assign IP addresses	Manage user sessions	Encrypt data	Manage the flow of data packets
49	In which situation would you use a repeater?	To connect different networks	To extend signal range	To manage network traffic	To secure data	To extend signal range
50	Which network topology is typically used in a wireless LAN?	Star	Ring	Mesh	Bus	Star
51	What does the acronym "MAC" stand for in networking?	Media Access Control	Maximum Access Control	Medium Access Channel	Managed Access Control	Media Access Control
52	What type of addressing does IPv4 use?	128-bit	64-bit	32-bit	16-bit	32-bit
53	What type of addressing does IPv6 use?	128-bit	64-bit	32-bit	16-bit	128-bit
54	What is the main function of a proxy server?	Store data	Filter requests	Manage traffic	Connect networks	Filter requests
55	What is the parity bit for the binary number 1011001 using even parity?	0	1	2	3	0
56	For the binary number 1100110, what parity bit is needed for odd parity?	0	1	2	3	1
57	If the data is 101001 and you want even parity, What is the required total number of 1s?	1	2	3	4	4
58	Which of the following is a subnet mask for C class IP address?	255.0.0.0	255.255.0.0	255.255.255.0	255.255.255.255	255.255.255.0
59	Which class does the IP address 172.16.0.1 belong to?	Class A	Class B	Class C	Class D	Class B
60	Which class does the IP address 10.0.0.5 belong to?	Class A	Class B	Class C	Class D	Class A
61	Which class does the IP address 192.168.15.10 belong to?	Class A	Class B	Class C	Class D	Class C
62	What does CIDR stand for?	Classful Inter-Domain Routing	Classless Inter-Domain Routing	Common Inter-Domain Routing	Complex Inter-Domain Routing	Classless Inter-Domain Routing
63	What is the primary benefit of CIDR over traditional IP addressing?	Simplicity in routing tables	Efficient use of IP addresses	Increased network speed	Better security	Efficient use of IP addresses
64	In CIDR, what does a prefix of /24 signify?	16 hosts	256 hosts	512 hosts	128 hosts	256 hosts
65	Which of the following are the network services?	File service	Print service	Database service	All of the above	All of the above
66	The performance of data communications network depends on	Number of users	The hardware and software	The transmission	All of the above	All of the above
67	Which of the communication modes support two way traffic but in only once direction of a time?	Simplex	Half-duplex	Full duplex	All of the above	Half-duplex
68	Which mode of communication allows data to be transmitted in both directions simultaneously?	Simplex	Half-duplex	Full-duplex	Unidirectional	Full-duplex
69	In which communication mode is data transmitted in one direction only?	Simplex	Half-duplex	Full-duplex	Bidirectional	Simplex
70	Which of the following is an example of simplex communication?	Radio broadcast	Phone call	Video conference	Instant messaging	Radio broadcast
71	Which of the following is not a mechanism that DHCP supports for IP address allocation?	Automatic allocation	Static allocation	Dynamic allocation	Faster allocation	Static allocation

72	What is the primary goal of cryptography?	Data compression	Data integrity	Data encryption	Data visualization	Data encryption
73	Which of the following is NOT considered a security goal?	Confidentiality	Integrity	Availability	Complexity	Complexity
74	Which type of cryptographic algorithm uses the same key for both encryption and decryption?	Asymmetric	Symmetric	Hashing	Salting	Symmetric
75	What service ensures that data has not been altered during transmission?	Authentication	Non-repudiation	Integrity	Confidentiality	Integrity
76	Which security service provides proof of the origin of data?	Encryption	Authentication	Integrity	Availability	Authentication
77	What is the main purpose of HTTP in the context of the World Wide Web?	Email transmission	File transfer	Web page retrieval	Secure transactions	Web page retrieval
78	What does WWW stand for?	World Wide Web	Wide World Web	Web World Wide	World Web Wide	World Wide Web
79	Which congestion control algorithm uses a fixed rate for output regardless of input burstiness?	Token Bucket	Leaky Bucket	Load Shedding	Random Early Detection	Leaky Bucket
80	In the token bucket algorithm, what happens when a token is not used within a certain time frame?	It is discarded	It is stored indefinitely	It is converted to credits	It is doubled	It is discarded
81	What is the main purpose of load shedding in congestion control?	To increase throughput	To reduce network traffic	To prioritize critical packets	To enhance data integrity	To reduce network traffic
82	Which algorithm allows bursts of data to be sent up to a maximum defined rate but controls the average rate?	Leaky Bucket	Token Bucket	Both A and B	None of the above	Token Bucket
83	How does the leaky bucket algorithm handle excess traffic?	It allows overflow	It queues packets	It drops excess packets	It sends them later	It drops excess packets
84	What is the primary function of the Reverse Address Resolution Protocol (RARP)?	To map IP addresses to MAC addresses	To map MAC addresses to IP addresses	To provide dynamic IP addresses	To resolve hostnames	To map MAC addresses to IP addresses
85	Which layer of the OSI model does RARP operate at?	Application Layer	Transport Layer	Network Layer	Data Link Layer	Network Layer
86	What is the maximum theoretical efficiency of the pure Aloha protocol?	18.40%	36.80%	50%	100%	18.40%
87	In the context of Aloha, what happens when a collision occurs?	The packet is immediately retransmitted	The packet is dropped and not retransmitted	A random backoff time is used before retransmission	The sender waits for an acknowledgment	A random backoff time is used before retransmission
88	In the Stop and Wait ARQ protocol, what happens after a sender transmits a frame?	It sends multiple frames	It waits for an acknowledgment	It sends the next frame immediately	It retransmits the frame	It waits for an acknowledgment
89	What is the maximum number of outstanding frames in the Go-Back-N ARQ protocol?	1	N	N-1	2	N
90	Which protocol allows the receiver to accept frames out of order?	Stop and Wait	Go-Back-N	Selective Repeat	Piggybacking	Selective Repeat
91	In the Sliding Window protocol, what does the window size determine?	Number of retransmissions	Number of frames that can be sent before an acknowledgment	Maximum delay allowed	Frame size	Number of frames that can be sent before an acknowledgment
92	What technique does piggybacking use to improve efficiency in data transmission?	Sending frames without acknowledgment	Combining data and acknowledgment frames	Using multiple channels	Encrypting data	Combining data and acknowledgment frames
93	Which of the following is true about Go-Back-N ARQ?	It can only send one frame at a time	It can resend all frames after a lost frame	It does not require acknowledgments	It accepts frames out of order	It can resend all frames after a lost frame
94	In Selective Repeat ARQ, how does the sender handle lost frames?	It ignores them	It retransmits all frames	It retransmits only the lost frames	It waits for the next acknowledgment	It retransmits only the lost frames
95	Which protocol is characterized by having a fixed window size that allows multiple frames to be sent?	Stop and Wait	Go-Back-N	Selective Repeat	Sliding Window	Sliding Window

96	What is the primary disadvantage of the Stop and Wait protocol?	Low reliability	High latency	Inefficient bandwidth usage	Complexity	Inefficient bandwidth usage
97	Which IEEE 802 standard is associated with wireless LANs?	IEEE 802.3	IEEE 802.11	IEEE 802.15	IEEE 802.16	IEEE 802.11
98	What does the IEEE 802.3 standard define?	Wireless networking	Ethernet networking	Bluetooth technology	Personal Area Networking	Ethernet networking
99	Which feature allows users to visualize network traffic in Cisco Packet Tracer?	Simulation mode	Real-time mode	Configuration mode	Design mode	Simulation mode
100	What does the "Physical" view in Cisco Packet Tracer represent?	Logical layout of the network	Actual physical arrangement of devices	Network protocol operations	Packet transmission details	Actual physical arrangement of devices
101	Which version of IP provides a larger address space?	IPv4	IPv5	IPv6	IPv7	IPv6
102	What is the primary purpose of the IP protocol?	Error detection	Reliable data delivery	Addressing and routing data	Data encryption	Addressing and routing data
103	In an IP address, what does the term "subnet mask" refer to?	The total number of hosts in a network	The range of IP addresses available	The portion that identifies the network	The type of routing protocol used	The portion that identifies the network
104	What is the primary purpose of CSMA/CA?	To minimize collisions in data transmission	To increase data transmission speed	To enhance error detection	To manage network congestion	To minimize collisions in data transmission
105	In CSMA/CA, what action does a device take before transmitting data?	Wait for a fixed time	Send data immediately	Sense the carrier	Transmit data blindly	Sense the carrier
106	What is the primary purpose of error detection techniques?	To correct data errors	To identify data errors	To improve transmission speed	To reduce data size	To identify data errors
107	What does a Hamming code provide in addition to error detection?	Error correction	Data compression	Data encryption	Network security	Error correction
108	Which error detection technique involves appending a fixed number of bits based on the count of 1s in a message?	Checksum	Parity bit	CRC	Hamming code	Parity bit
109	In the context of CRC, what does the term "divisor" refer to?	The data being transmitted	The generator polynomial	The receiver's checksum	The sender's address	The generator polynomial
110	What is the main purpose of modulation in communication systems?	To increase bandwidth	To convert signals to a suitable form for transmission	To enhance security	To compress data	To convert signals to a suitable form for transmission
111	Which of the following is an example of analog modulation?	Frequency Shift Keying	Amplitude Modulation	Phase Shift Keying	Amplitude Shift Keying	Amplitude Modulation
112	In digital modulation, what does the term "baud rate" refer to?	The number of bits transmitted per second	The number of signal changes per second	The frequency of the carrier wave	The length of the message	The number of signal changes per second
113	Which modulation technique varies the amplitude of the carrier signal based on the data signal?	Frequency Modulation	Amplitude Modulation	Phase Modulation	Frequency Shift Keying	Amplitude Modulation
114	Which modulation technique is less susceptible to noise and interference?	Frequency Modulation	Amplitude Modulation	Phase Modulation	Frequency Shift Keying	Frequency Modulation
115	Which type of unguided media is commonly used for broadcasting audio and television signals?	Microwave	Infrared	Radio	Fiber optic	Radio
116	What is a primary characteristic of microwave transmission?	It requires a clear line of sight	It can penetrate walls easily	It operates over short distances	It uses sound waves	It requires a clear line of sight
117	Which unguided media is typically used for short-range communication, such as remote controls?	Radio	Infrared	Microwave	Fiber optic	Infrared
118	What frequency range do microwaves typically operate in?	Below 3 kHz	3 GHz to 30 GHz	30 kHz to 300 kHz	Above 300 GHz	3 GHz to 30 GHz
119	Which of the following is a characteristic of static routing?	Automatically adjusts to network changes	Requires manual configuration	Uses routing algorithms	Supports load balancing	Requires manual configuration

120	What is the main advantage of dynamic routing protocols over static routing?	Simplicity	Lower overhead	Automatic updates	Greater security	Automatic updates
121	What is the primary purpose of subnetting?	To increase bandwidth	To improve network security	To divide a larger network into smaller segments	To simplify routing	To divide a larger network into smaller segments
122	If you have the IP address 192.168.1.10 and a subnet mask of 255.255.255.0, what is the network address?	192.168.1.0	192.168.1.10	192.168.1.255	192.168.0.0	192.168.1.0
123	How many subnets can be created from a Class C address with a default subnet mask of 255.255.255.0 if 2 bits are borrowed?	2	4	8	16	4
124	What is the broadcast address for the subnet 192.168.10.0 with a subnet mask of 255.255.255.128?	192.168.10.0	192.168.10.127	192.168.10.255	192.168.10.64	192.168.10.127
125	Which of the following is a key difference between TCP and UDP?	TCP is connectionless; UDP is connection-oriented	TCP provides error correction; UDP does not	TCP is faster than UDP in all cases	UDP guarantees delivery; TCP does not	TCP provides error correction; UDP does not
126	What type of flow control mechanism does TCP use?	Stop-and-wait	Sliding window	Rate-based	None	Sliding window
127	In TCP, what is the purpose of the three-way handshake?	To establish a connection	To terminate a connection	To synchronize data streams	To ensure error detection	To establish a connection
128	Which field in the TCP header is used to indicate the sequence of packets?	Acknowledgment number	Sequence number	Checksum	Window size	Sequence number
129	In which scenario would you prefer to use UDP over TCP?	For file transfers	For real-time audio/video streaming	For web browsing	For email transmission	For real-time audio/video streaming
130	In a checksum calculation, what happens if the result equals zero?	The data is corrupted	The data is valid	The checksum is discarded	The data is encrypted	The data is valid
131	How many parity bits are needed to detect and correct a single-bit error in a 7-bit data word using Hamming code?	1	2	3	4	3
132	Which type of cable uses an insulating layer to separate the inner conductor from the outer conductor?	Twisted pair cable	Coaxial cable	Fiber optic cable	Copper cable	Coaxial cable
133	In which scenario would fiber optic cables be preferred over twisted pair cables?	Short-distance LAN connections	Environments with high electromagnetic interference	Low-cost residential wiring	Temporary installations	Environments with high electromagnetic interference
134	What advantage do fiber optic cables have over coaxial cables in terms of signal loss?	Higher signal attenuation	Lower signal attenuation	No signal loss	Higher susceptibility to interference	Lower signal attenuation
135	Which of the following technologies allows for the creation of virtual networks on a physical network?	LAN	VLAN	VPN	NAT	VLAN
136	What is a computer network?	A system for storing data	A collection of interconnected computers that share resources	A type of software application	A method for encrypting data	A collection of interconnected computers that share resources
137	What is data communication?	The exchange of data between devices	The physical connection of cables	The process of storing data	The encryption of messages	The exchange of data between devices
138	What does latency refer to in data communication?	The speed of the network	The time delay in data transmission	The amount of data lost during transmission	The physical distance between devices	The time delay in data transmission
139	What is the primary purpose of a network protocol?	To store data securely	To define rules for data communication	To increase data speed	To encrypt network traffic	To define rules for data communication
140	Which of the following is a feature of digital communication?	Continuous signals	Discrete signals	High susceptibility to noise	Lower transmission efficiency	Discrete signals
141	What is the Internet?	A private network	A global network of interconnected computers	A local area network	A type of software application	A global network of interconnected computers
142	What protocol is primarily used for transmitting web pages over the Internet?	FTP	SMTP	HTTP	SNMP	HTTP
143	How many layers does the OSI model have?	4	5	6	7	7

144	Which model is primarily used for internet communications?	OSI model	TCP/IP model	Both models equally	Neither model	TCP/IP model
145	What does the ping command primarily test?	Network bandwidth	IP address configuration	Reachability of a host	Firewall settings	Reachability of a host
146	Which of the following will the ipconfig command display?	Current network connections	IP address and subnet mask	Network traffic statistics	Active processes	IP address and subnet mask
147	What is the purpose of the ping command's TTL (Time to Live) value?	To determine the maximum data transfer rate	To limit the number of hops a packet can take	To measure the response time	To configure network settings	To limit the number of hops a packet can take
148	Which of the following is/are components of data communication?	Message	Sender	Receiver	All of the above	All of the above
149	What is the role of the receiver in the data communication process?	To send data	To encrypt data	To decode and interpret the message	To transmit data	To decode and interpret the message
150	Which component of data communication refers to the medium used to transmit the message?	Protocol	Channel	Encoder	Decoder	Channel
151	What is the primary format used for representing data in digital networks?	Analog signals	Binary data	ASCII characters	Text files	Binary data
152	Which of the following best describes "data flow" in a network?	The movement of data between devices	The encryption of data	The compression of data	The storage of data	The movement of data between devices
153	In the context of networks, what does "throughput" refer to?	The amount of data received	The maximum data transfer capacity	The time taken to transmit data	The error rate	The maximum data transfer capacity
154	Which one of following represent IPV4 loopback address?	255.255.0.0	127.0.0.0	127.0.0.1	255.0.0.1	127.0.0.1
155	Which class IP addresses are reserved for special use?	Class A	Class B	Class C	Class D	Class D
156	Which of the following is not a reason of congestion in network?	Low bandwidth	Multicasting	Few hosts in broadcast domain	Outdated hardware	Few hosts in broadcast domain
157	Which of the following is not the layer of TCP/IP protocol?	Application Layer	Transport Layer	Session Layer	Internetwork layer	Session Layer
158	Fullform of SSL is ?	Security Subscription Layer	Secure Socket Layer	Solid Secure Layer	Secure Service Layer	Secure Socket Layer
159	What is the primary function of Wireshark?	To configure network devices	To analyze network traffic	To create network diagrams	To encrypt data	To analyze network traffic
160	Which of the following features does Wireshark provide?	Real-time data encryption	Packet filtering and analysis	Bandwidth monitoring	Firewall protection	Packet filtering and analysis
161	Which technique is used to reduce congestion in a network?	Token passing	CSMA/CD	ALOHA	Sliding window	Token passing
162	What does the term "bandwidth" refer to in networking?	Distance of signal	Maximum data rate	Signal strength	Packet size	Maximum data rate
163	What is the primary purpose of a MAC address?	Identifying devices	Encrypting data	Managing sessions	Routing packets	Identifying devices
164	What is the main purpose of a digital certificate?	Data compression	Identity verification	Packet routing	Session management	Identity verification
165	Which topology provides the highest redundancy?	Star	Mesh	Bus	Ring	Mesh
166	What is the main purpose of the ICMP protocol?	Error reporting	File transfer	Email communication	Remote access	Error reporting
167	What is a primary disadvantage of a star topology?	High cost	High redundancy	Limited scalability	Single point of failure	Single point of failure

168	What does the acronym BGP stand for in networking?	Border Gateway Protocol	Binary Gate Protocol	Basic General Protocol	Bandwidth Gateway Protocol	Border Gateway Protocol
169	What is the role of the control plane in a router?	Forwarding packets	Managing routing protocols	Filtering data	Encrypting traffic	Managing routing protocols
170	Which of the following is a class of IP address reserved for multicast?	Class A	Class B	Class C	Class D	Class D
171	Which device converts digital signals to analog signals?	Router	Switch	Modem	Hub	Modem
172	In a network, what does the term “load balancing” refer to?	Distributing traffic evenly	Encrypting data	Managing IP addresses	Establishing connections	Distributing traffic evenly
173	What does the term "overhead" refer to in networking?	Extra data for management	Data lost during transfer	Packet size	Number of connections	Extra data for management
174	Which of the following is an example of a wide area network (WAN)?	Home network	Campus network	The Internet	Local area network	The Internet
175	What type of attack involves overwhelming a server with traffic?	Phishing	Spoofing	Denial of Service	Man-in-the-middle	Denial of Service
176	What type of addressing does IPv6 use?	Binary	Decimal	Hexadecimal	Alphanumeric	Hexadecimal
177	What type of network uses a central controller for communication?	Peer-to-peer	Client-server	Hybrid	Decentralized	Client-server
178	What is the primary purpose of firewalls in networks?	Packet analysis	Network segmentation	Security	Bandwidth management	Security
179	What type of protocol is ICMP?	Connection-oriented	Connectionless	Reliable	Stateful	Connectionless
180	In which scenario is a VPN primarily used?	Remote access	Local area networking	Wide area networking	Network backup	Remote access
181	Which of the following is NOT a characteristic of TCP?	Connection-oriented	Reliable	Flow control	Stateless	Stateless
182	Which topology is characterized by a loop?	Star	Ring	Bus	Mesh	Ring
183	What is the role of a subnet mask?	To identify the network part	To encrypt data	To compress data	To segment files	To identify the network part
184	Which of the following is a disadvantage of a bus topology?	High cost	Limited cable length	Difficult to troubleshoot	Single point of failure	Single point of failure
185	What does QoS stand for in networking?	Quality of Service	Quantity of Service	Query of Service	Quick Operating System	Quality of Service
186	What is the primary function of the transport layer?	Route packets	Ensure reliable transmission	Manage sessions	Handle error detection	Ensure reliable transmission
187	In a wireless network, what does SSID stand for?	Standard Service Identifier	Service Set Identifier	Security System Identifier	Signal Strength Identifier	Service Set Identifier
188	Which type of cable is commonly used in Ethernet networks?	Coaxial	Twisted pair	Fiber optic	Serial	Twisted pair
189	What is the primary role of the application layer?	Data transmission	User interface	Routing	Session management	User interface
190	Which protocol is primarily used for file transfers on the Internet?	SMTP	POP3	FTP	HTTP	FTP
191	What does VLAN stand for?	Virtual Local Area Network	Virtual Line Area Network	Variable Local Area Network	Virtual Link Area Network	Virtual Local Area Network

192	Which of the following addresses is Used for local communication within a private network?	127.0.0.1	8.8.8.8	172.16.0.1	10.0.0.1	10.0.0.1
193	In networking, what does the term "ping" refer to?	A network troubleshooting tool	A file transfer protocol	A routing protocol	A data compression method	A network troubleshooting tool
194	What is the main advantage of using fiber optic cables?	Low cost	High bandwidth	Easy installation	Greater signal loss	High bandwidth
195	What does the acronym ISP stand for?	Internet Service Provider	Internal Service Provider	Internet Service Provider	Internal Service Provider	Internet Service Provider
196	What protocol is used for network time synchronization?	SNTP	FTP	HTTP	SMTP	SNTP
197	Which of the following is NOT a valid IP address?	192.168.0.1	256.100.50.25	10.0.0.1	172.16.0.5	256.100.50.25
198	In TCP, what does the term “window” refer to?	Data size	Buffer size	A type of packet	Transmission speed	Buffer size
199	Which of this is not a network edge device?	PC	Smartphone	Switch	Laptop	Switch
200	What was the name of the first network?	CNNET	ASAPNET	ARPANET	NSFNET	ARPANET